KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
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SDC 1-Day 9

II-II CSE, CSE(AI&ML)
AY:2026-2027

SL No	Week	Part A	Detailed Subtopics	Part B	Detailed Subtopics
1	Week 4	Unsupervised algorithms	Introduction to Unsupervised Learning, Types of Unsupervised Learning, Cluster Analysis, K-Means Clustering, How Does the K-Means Algorithm Work?, Elbow Method, Limitations of K-Means, Solutions to the Limitations,	Practical Implementation	Day9_Exercise8_KMeans_MallCustomers

Contents:

1.Introduction to Unsupervised Learning

2. Types of Unsupervised Learning

3. Cluster Analysis

4. K-means Clustering

5. How does the K-Means Algorithm Work?

6.Elbow Method

7. Limits of K-Means

8. Solutions to Limitations:

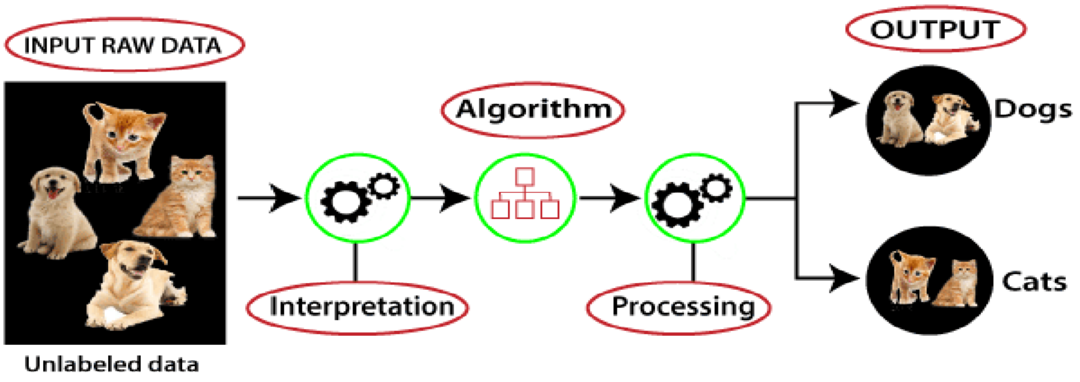
9.Q & A

1.Introduction to Unsupervised Learning

- Unsupervised learning algorithms are used when we don’t have labeled data.

  These models try to identify hidden structures or patterns in the data.

- The main objective is to explore the data and find inherent groupings or relationships.



A) Supervised Learning: Like a student learning with a teacher providing answers and feedback.
B) Unsupervised Learning: Like a student discovering patterns and relationships on their own without answers provided.

Example:

Suppose a company has information about thousands of customers, including their:

- Age
- Income
- Spending behaviour
- Purchase history

But the company does not know how these customers can be categorized.

For example, the company may want to identify whether there are naturally occurring groups such as:

- Customers who have high income and spend a lot
- Customers who have high income but spend very little
- Customers with low income and low spending
- Customers who frequently purchase certain types of products

However, these categories are **not already available in the dataset**.

The company only has the customer information.

Now the objective is not to predict a known output. Instead, the objective is to analyze the available data and answer questions such as:

Are there groups of customers with similar behaviour?

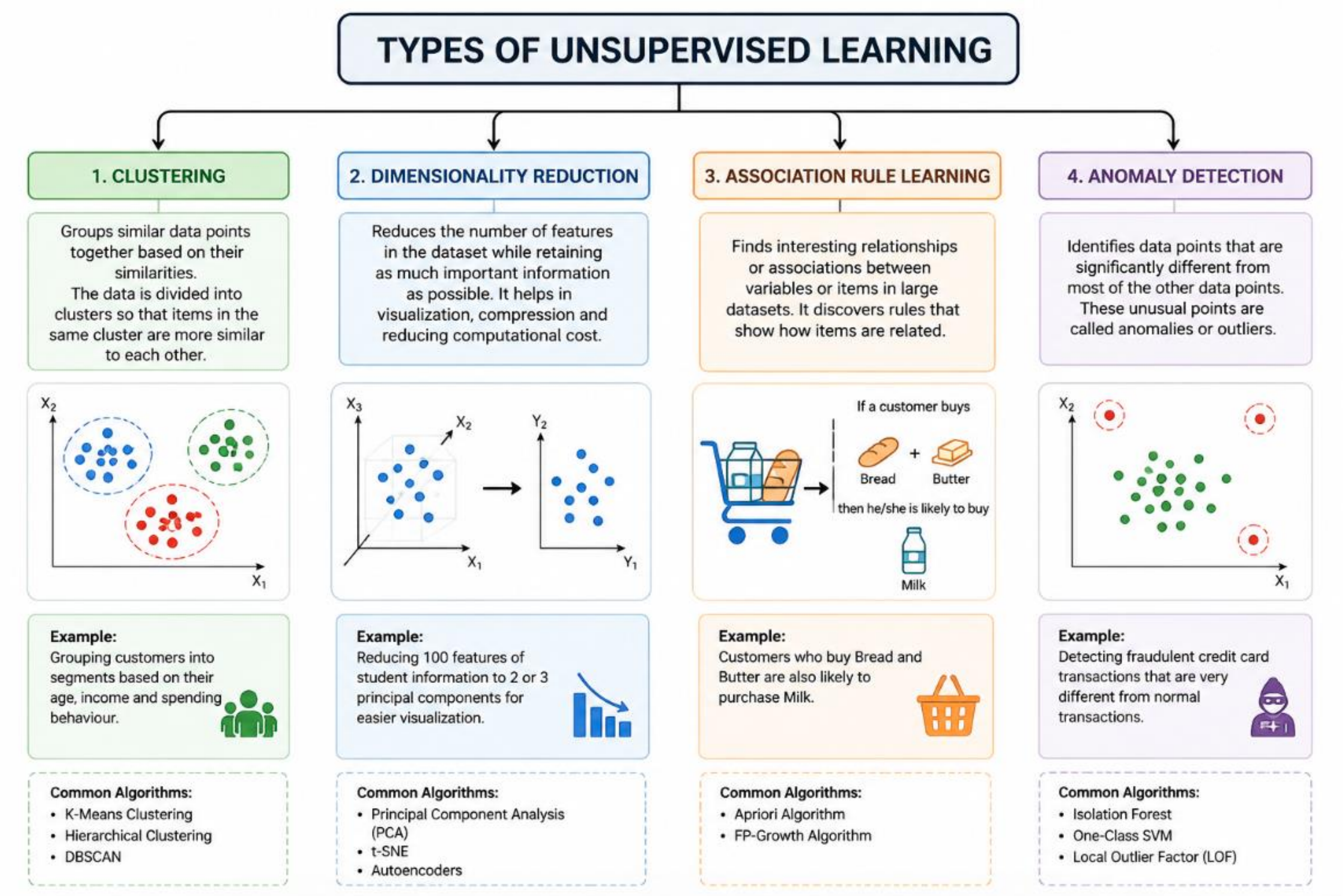
Can we identify hidden patterns in customer data?

Are some customers similar to each other based on their characteristics?

In this situation, the machine learning algorithm is not provided with the correct categories or answers. Instead, it analyzes the data and tries to discover meaningful patterns or groups on its own.

This type of learning is called **Unsupervised Learning**.

2. Types of Unsupervised Learning

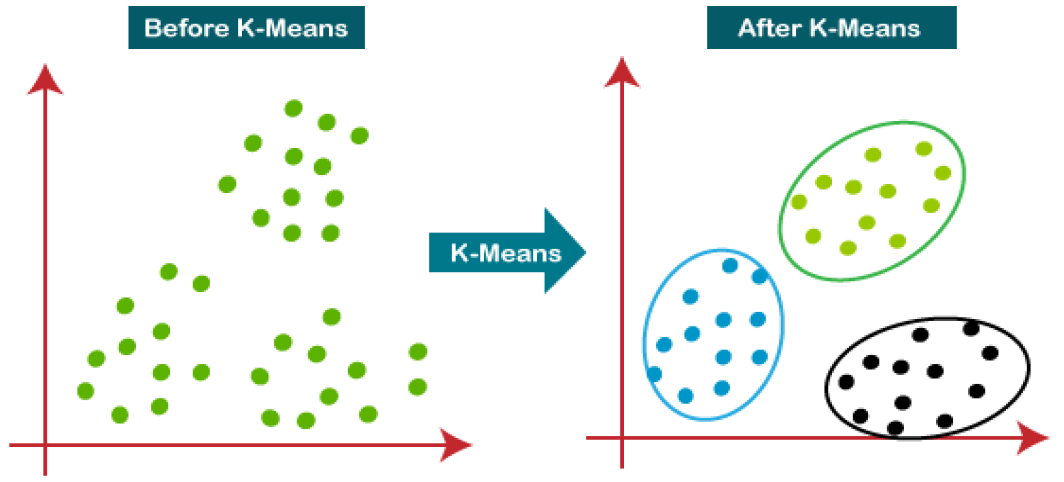


3. Cluster Analysis

In unsupervised machine learning, cluster analysis groups unlabeled input data into meaningful output data. This output data can be used to understand the structure of the input data better or to make predictions about unlabeled data. There are many different ways to perform cluster analysis, but the most common method is to use a model that groups input data based on similarity.

4.K-means Clustering

Definition: K-Means is one of the most widely used clustering algorithms in unsupervised learning. It aims to partition a dataset into K clusters, where each data point belongs to the cluster with the nearest mean (centroid). The goal of K-Means is to minimize the variance within each cluster.



5.How does the K-Means Algorithm Work?

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

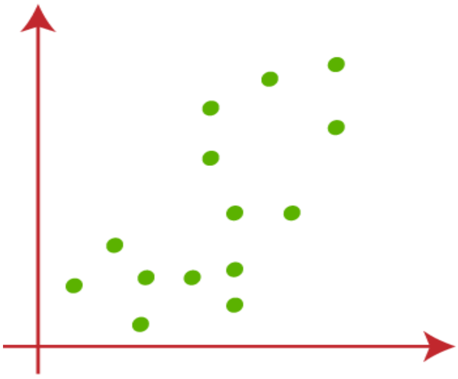
Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

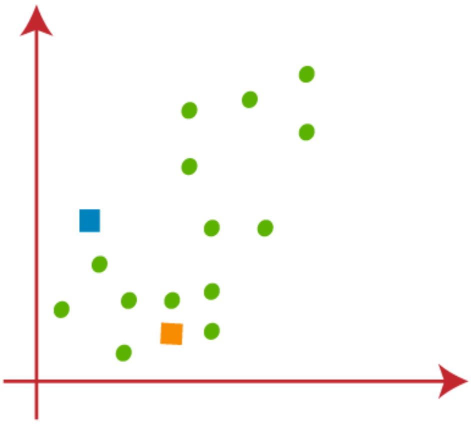
Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

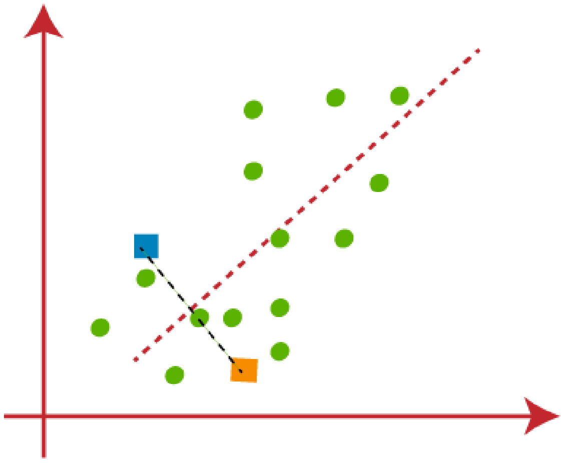
Suppose we have two variables M1 and M2. The x-y axis scatter plot of these two variables is given below:



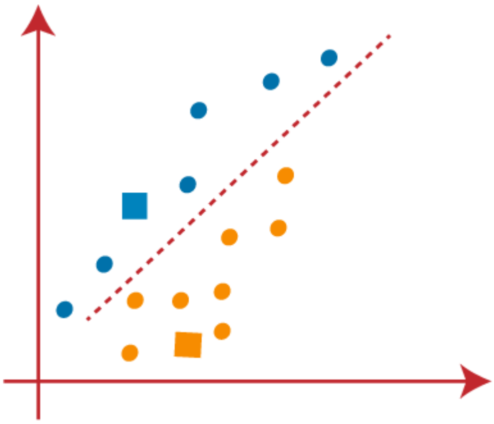
- Let's take number k of clusters, i.e., K=2, to identify the dataset and to put them into different clusters. It means here we will try to group these datasets into two different clusters.
- We need to choose some random k points or centroid to form the cluster. These points can be either the points from the dataset or any other point. So, here we are selecting the below two points as k points, which are not the part of our dataset.
  Consider the below image:



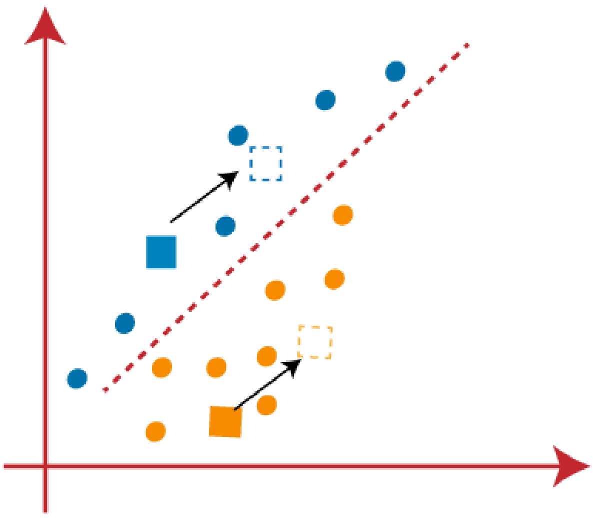
- Now we will assign each data point of the scatter plot to its closest K-point or centroid. We will compute it by applying some mathematics that we have studied to calculate the distance between two points. So, we will draw a median between both the centroids. Consider the below image:



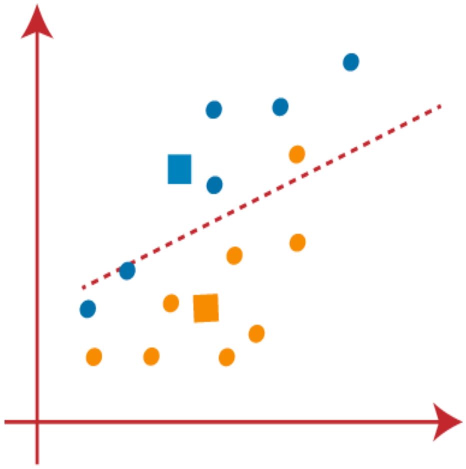
From the above image, it is clear that points left side of the line is near to the K1 or blue centroid, and points to the right of the line are close to the yellow centroid. Let's color them as blue and yellow for clear visualization.



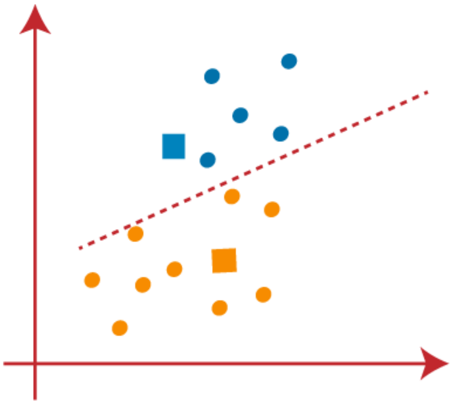
- As we need to find the closest cluster, so we will repeat the process by choosing **a ne centroid**. To choose the new centroids, we will compute the center of gravity of these centroids, and will find new centroids as below:



- Next, we will reassign each datapoint to the new centroid. For this, we will repeat the same process of finding a median line. The median will be like below image:

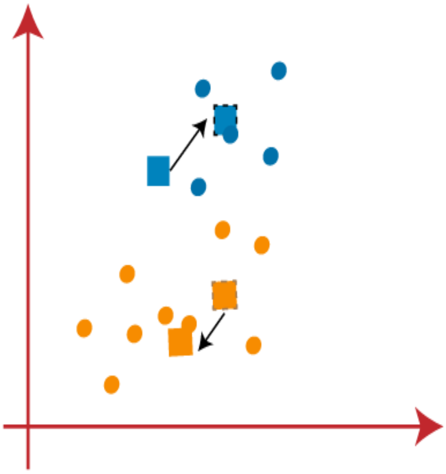


From the above image, we can see, one yellow point is on the left side of the line, and two blue points are right to the line. So, these three points will be assigned to new centroids.

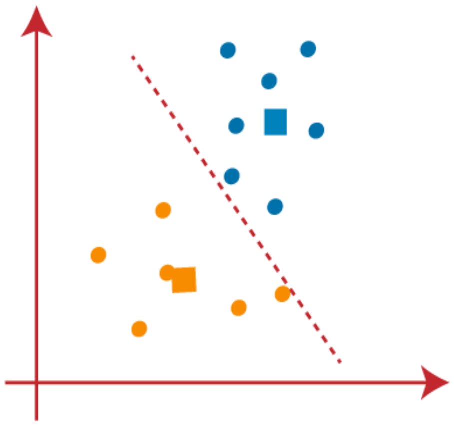


As reassignment has taken place, so we will again go to the step-4, which is finding new centroids or K-points.

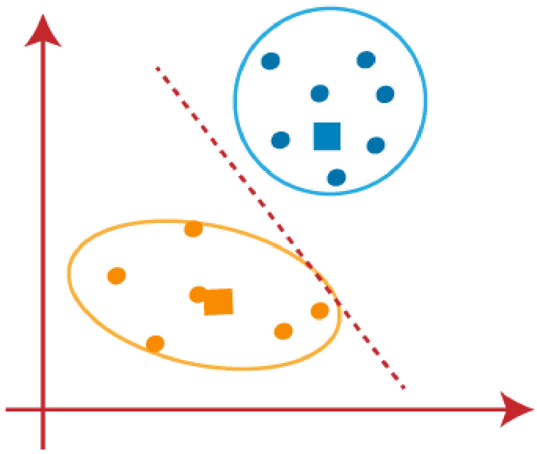
- We will repeat the process by finding the center of gravity of centroids, so the new centroids will be as shown in the below image:



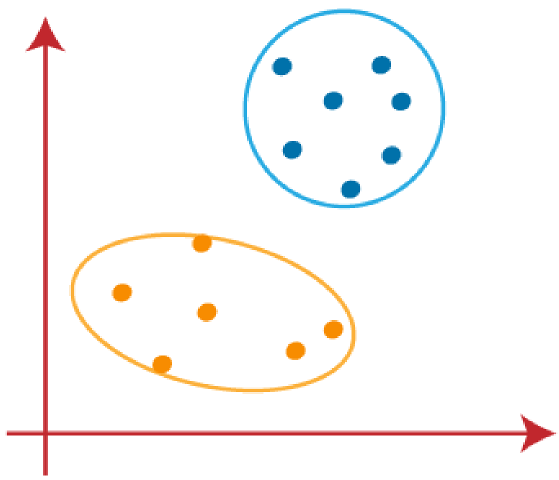
- As we got the new centroids so again will draw the median line and reassign the data points. So, the image will be:



- We can see in the above image; there are no dissimilar data points on either side of the line, which means our model is formed. Consider the below image:

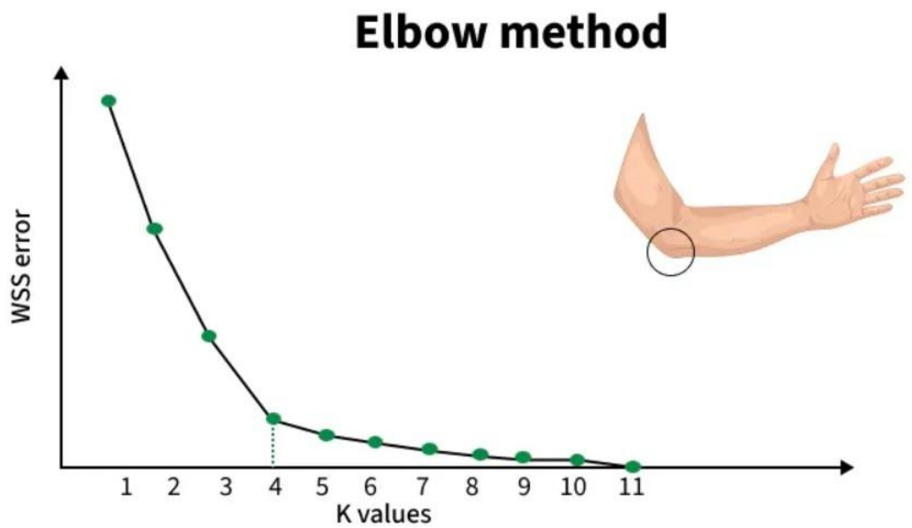


As our model is ready, so we can now remove the assumed centroids, and the two final clusters will be as shown in the below image:



6. Elbow Method

How to choose the value of "K number of clusters" in K-means Clustering?
In K-Means clustering, the algorithm partitions data into k clusters by minimizing the distances between points and their cluster centroids. However, deciding the ideal k is not straightforward. The Elbow Method helps by plotting the Within-Cluster Sum of Squares (WCSS) against increasing k values and looking for a point where the improvement slows down, this point is called the "elbow."



Working of Elbow Point

The Elbow Method works in the following steps:

1. We begin by selecting a range of k values (for example, 1 to 10).

2. For each k, we run K-Means and calculate WCSS (Within-Cluster Sum of Squares), which shows how close the data points are to their cluster centroids:

$$\boxed{WCSS = \sum_{i=1}^{K} \sum_{x \in C_i} ||x - \mu_i||^2}$$

Where:

- K= Total number of clusters
- C_i= The i^{th}cluster
- x= A data point belonging to that cluster
- μ_i= Centroid of the i^{th}cluster
- $|| x - \mu_i ||^2$= Squared distance between the data point and its cluster centroid

3. After computing WCSS for all k values, we plot k vs WCSS.
4. WCSS always decreases as k increases because more clusters reduce the internal spread.

5. However, after a certain point, the improvement becomes very small. This bend or “elbow” in the curve indicates the point where adding more clusters no longer gives meaningful improvement.

- **Before the elbow:** WCSS drops quickly -> clusters become much better.
- **After the elbow:** WCSS drops slowly -> extra clusters add little value and may lead to overfitting.

The goal is to identify the point where the rate of decrease in WCSS sharply changes, indicating that adding more clusters (beyond this point) yields diminishing returns. This "elbow" point suggests the optimal number of
clusters.

7.Limits of K-Means

While K-Means is a simple and effective algorithm, it has several limitations:

1. Predefined K:

Limitation: The number of clusters (K) must be specified beforehand. If you don't know the number of clusters in advance, this can be a challenge.

2. Sensitivity to Initial Centroids:

Limitation: K-Means is sensitive to the initial placement of centroids. If the initial centroids are poorly chosen, it can lead to poor clustering results.

3. Sensitivity to Outliers:

Limitation: K-Means is sensitive to outliers since they can significantly influence the centroid position.

4. Convergence to Local Minima:

Limitation: K-Means can converge to local minima, meaning it might not find the best possible cluster configuration. Running the algorithm multiple times with different initializations can help mitigate this

8.Solutions to Limitations:

- **Elbow Method**: To determine the optimal K by looking for a "knee" or "elbow" in the inertia plot (sum of squared distances of points to their centroids).
- **K-Means++**: A smarter initialization technique that spreads out the initial centroids, improving convergence and results.
- **Use of other algorithms**: When K-Means is unsuitable, alternative clustering algorithms like DBSCAN (density-based) or hierarchical clustering can be used.

9.Q & A

1. A dataset visually appears to contain three natural groups, but K-Means is applied with $K = 2$. What is likely to happen?

Answer:
Some observations from different natural groups may be combined into the same cluster. This results in less distinct and more heterogeneous clusters.

2. If $K$is increased from 3 to 10, will WCSS increase or decrease? Why?

Answer:
WCSS will generally decrease because more centroids allow data points to be closer to their assigned centroids. However, this does not mean $K = 10$is automatically better.

3. K-Means produces different results when run twice on the same dataset. What could be the reason?

Answer:
The initial centroids may be selected differently in each run. Different starting positions can lead the algorithm to different final cluster configurations.

4. A dataset contains Age and Annual Income. Income has values in lakhs, while Age ranges from 20 to 60. What should be considered before applying K-Means?

Answer:
Feature scaling should be considered because K-Means uses distance. Without scaling, Income may dominate the distance calculation due to its larger numerical range.

5. Suppose one centroid stops changing, but another centroid continues to move. Has the algorithm converged?

Answer:
Not necessarily. K-Means is considered converged when the overall centroid positions or cluster assignments stop changing significantly.

6. If every data point becomes its own cluster, what will happen to WCSS?

Answer:

$$WCSS = 0$$

Each data point is its own centroid, so the distance between every point and its centroid is zero. However, such clustering is not meaningful.

7. Why can an outlier significantly affect K-Means clustering?

Answer:
K-Means calculates centroids using the mean. An extreme observation can pull the centroid toward itself, affecting the assignment of other nearby observations.

8. If $K = 1$, what does K-Means do?

Answer:
All observations are assigned to a single cluster. The centroid represents the mean position of the entire dataset.

9. If two initial centroids are placed very close to each other, what problem might occur?

Answer:
Both centroids may initially represent the same region of the data, while another region may not be represented properly. This can lead to a poor clustering solution.

10. Suppose adding a new feature changes the clusters completely. Does this mean K-Means is unstable?

Answer:
Not necessarily. The new feature changes the distance calculations between observations. If the feature contains important information, different clusters may be expected.